

As most of the users aren't in India, outside teams define what is needed next.

—Vivek Sharma, vice president-emerging markets region and director-India Design Center, STMicroelectronics



dia generated revenues of \$3.25 billion in 2005, and could reach \$14.42 billion by 2010, and further to \$43 billion by 2015, growing at a compounded annual growth rate (CAGR) of 30 per cent during the forecast period.

However, "given the prevailing slowdown, the ISA has revised the estimated growth rate for the calendar year 2009 to 13 per cent, which is roughly half of the growth rate exhibited before. In year 2006, the semiconductor industry in India had grown 25-30 per cent. And it had been growing annually at this rate for the last few years. So we clearly see the impact of recession on the local semiconductor market," informs Sharma.

Qualitatively, let's look at the local market from two perspectives. What we consume in India has two components:

1. Total market (TM) for semiconductors. All the semiconductors that we use in India irrespective of whether we are manufacturing them here or bring them from abroad
2. Total available market (TAM) for semiconductors. Semiconductors which are being used inside electronic items made in India, i.e., local semiconductor consumption

The data for year 2008 as per ISA report was \$6 billion for TM and \$2.5 billion for TAM in India. Whatever we are using in India, e.g., mobile phones, the semiconductor content that goes inside it is TM. If the mobile phones are manufactured here in India, the semiconductor content inside them is TAM.

Companies like LG are producing

designs in India not only for the local market but also for the world market. Most of the design centres aren't hiring any more, so surely there is impact on design centres which are designing for foreign markets.

The vast majority of the embedded processor market remains microcontroller units (MCUs), which are low-cost autonomous embedded processing units used in embedded applications having low-end processing requirements.

Embedded CPUs, DSPs and FPGAs all typically provide higher levels of processing performance than MCUs and command higher average selling prices, but exist in smaller unit quantities, making MCUs also the largest dollar-volume segment of the four embedded processing technologies.

Growth drivers

Demand-pull in the embedded space stems mainly from:

1. Consumer electronics
2. Security and surveillance including biometrics
3. Communication
4. Defense and avionics
5. Medical electronics, particularly portable monitoring devices

In consumer electronics, digital entertainment will be a high-growth area. Security and surveillance will see strong growth with increase in government spending. In medical electronics, developers are working on the latest ultrasound, MRI and diagnostic devices, for both local and overseas markets.

Other areas that will see growth

include automobiles (where we are seeing innovative products at breakthrough price points such as the Tata Nano), energy saving appliances and alternative energy devices like inverters and solar cells.

An ISA-IDC report observes that the power sector reforms in the Indian industry will help the embedded systems business with overall infrastructure upgrade and introduction of sophisticated electronics equipment for the industry.

Emerging trends

Consumer electronics applications are digital TV sets, digital camcorders, DVD players, set-top boxes, digital video recorders/home video servers, PDA cameras, network cameras, video phones, digital still cameras and smart phones.

In this area, we are seeing migration in technology from:

1. Fixed-function to multi-function devices
2. Analogue to digital media
3. Proprietary to commercial software
4. 8- and 16-bit to 32+ bit processors
5. Limited, fixed-line bandwidth to high-bandwidth wired/wireless lines

Processing needs range from high-performance multi-core processors for applications like base stations and medical image processing to energy-harvesting, embedded processors for implantable medical devices.

Another growing trend is the move towards small form-factor designs which can be upgraded with minimal application efforts. 'Small' and 'low power' are the buzzwords. The embedded solution providers are constantly innovating to match these requirements. For instance, the MSP430 platform of ultra-low-power 16-bit RISC mixed-signal processors from Texas Instruments (TI) is good for battery-powered measurement applications, while its OMAP platform delivers a variety of high-performance application processors with extremely